Arrays and selectors: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01 Half-selected voltages

A 2×2 array uses fixed V/2 bias: selected row R1 at 1.2 V, selected column C1 at 0 V, and both unselected lines 0.6 V. Find row-minus-column voltage at each cell and identify half-selected cells. Is 0.6 V automatically safe?

E02 Floating sneak path

In a 2×2 array, selected row R1 is 1 V and selected column C1 is 0 V; both other lines float. All cells are linear, wires ideal, the selected conductance is 0.1 µS and the other three are 1 µS. Find direct, three-cell sneak and total sensed currents, and both floating node potentials.

E03 Nonlinearity ratio

For one selector state, polarity and measurement timing, define $NL = I(1.2V)/I(0.6V)$. Assume $NL = 10^3$ and $I(1.2V) = 10\,\mu A$. Find $I(0.6V)$. Why does this not determine array read margin?

E04 Series equivalent circuit

Under a fixed-state linear approximation, selector resistance is 2 kΩ and memory resistance 4 kΩ, with 1.2 V applied across both. Find current and voltage on each device. Can fixed-resistance division be used for a nonlinear selector?

E05 Access structure and polarity

Compare 1R, 1T1R, 1S1R and 1D1R. For a memory requiring opposite SET and RESET biases, explain why diode orientation and the two operating windows must be checked.

E06 3D density and manufacturing

An ideal planar cell has footprint $4F^2$, with $F = 20$ nm. Assume each cell stores 1 bit and each of three perfectly stacked layers has the same number of cells and unchanged footprint. For this model, find ideal projected area per bit; list at least three reasons actual chip density differs.

Arrays and selectors: Exercises (continued)

E07 Self-selection and the sensing circuit

Assume a selected LRS cell conducts $10\ \mu$A at V_{read} and 10 nA at half bias; the selected HRS conducts $0.2\ \mu$A at V_{read}. A sense node is ideally clamped at 0 V and 99 other branches are each in LRS at half bias; each contributes 10 nA, unchanged when the selected state changes. Calculate the LRS nonlinearity, total unselected current and sensed HRS/LRS currents. Does this single nonlinearity determine the leakage of an arbitrary array?

E08 Passing sites and complete-array yield

For a model with 100 contacts, suppose each independently passes a specified test with probability 0.99, and there is no repair. Calculate the probability that all 100 pass and the expected number that pass. Which result requires independence? Does passing a contact test prove the memory array works?

E09 Forming and addressing a pattern

Compare AAO, nanosphere lithography and block-copolymer directed self-assembly: for each, describe how a pattern forms, how it can be transferred, and what must still be checked before it is electrically addressed. Explain one way defects or registration error can make a regular pattern unusable.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study